João Eduardo Pereira dos Santos

Senior Python Engineer | Backend, Cloud & AI/LLM Applications

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PROFESSIONAL SUMMARY

Senior Python Engineer with **16+ years of software engineering experience** building resilient backend services, REST APIs, enterprise integrations, and data processing solutions. Strong background in **Python, SQL**, relational databases, and cloud-based applications. Currently expanding into AI Engineering through hands-on work with **LLMs, Retrieval-Augmented Generation (RAG)**, embeddings, and AI agents. Experienced in delivering maintainable enterprise solutions across **Critical TechWorks, EY, and UNICAMP**.

TECHNICAL SKILLS

Backend Engineering: Python, FastAPI, Flask, Django, REST APIs, AsyncIO, Microservices, Clean Architecture

AI & LLM Applications: LLM Integration, Retrieval-Augmented Generation (RAG), Document Ingestion & Chunking, Embeddings, Semantic Retrieval, Prompt Engineering, AI Agents, Vector Databases

Databases & Data Engineering: PostgreSQL, Oracle DB, MySQL, SQL Server, SQL Optimization, ETL / ELT, IBM DataStage, SAP Migration

Cloud, DevOps & Practices: AWS, Docker, Kubernetes, Git, CI/CD, Jenkins, Linux, Unit Testing, Agile / Scrum

PROFESSIONAL EXPERIENCE

Python Full Stack Developer • **Critical TechWorks (BMW Group)** — Nov 2025 – Present | Porto, Portugal

- Architect and integrate Python backend microservices, REST APIs, and automotive data integration workflows.
- Deploy and maintain containerized services using Docker and Kubernetes within AWS cloud environments.
- Implement automated CI/CD pipelines, regression test suites, and agile delivery practices (Scrum).
- Optimize complex SQL queries, data access layers, and enterprise system connectivity for high availability.
- Collaborate with international engineering teams across Portugal and Germany delivering reliable software solutions.

Python Full Stack Developer • **Ernst & Young (EY)** — Dec 2022 – Nov 2025 | Brazil / Hybrid

- Developed a Python-based Retrieval-Augmented Generation (RAG) prototype for semantic search and Q&A over internal document sets.
- Engineered document ingestion, chunking pipelines, vector embeddings, and LLM query-response workflows.
- Delivered enterprise data integration pipelines for SAP migration projects, including extraction, transformation, and validation.
- Developed and maintained ETL workflows using IBM DataStage and custom Python data processing scripts.
- Authored complex SQL transformations, automated data validation routines, and operational reporting dashboards.

Senior Systems Engineer & Python Developer • **UNICAMP** — Apr 2008 – Dec 2022 (14+ yrs) | Brazil

- Engineered scalable web applications, RESTful APIs, and administrative platforms using Python (Django, Flask) and SQL.
- Architected and delivered core administrative and academic portal backends serving over 40,000 university users.
- Designed, optimized, and administered mission-critical relational databases (PostgreSQL, Oracle, MySQL).
- Spearheaded digital transformation initiatives, DevOps automation, and Linux server infrastructure management.
- Automated complex institutional data extraction, BI reporting pipelines, and routine operational workflows.

Earlier Career: *Digital Games Programming Instructor* at SESC Santana (2016–2019; Python, Lua, game logic) • *Software & Data Analyst* at Mercer Brasil (2005–2008; SQL Server, ETL, database integration) • *Technical Support* at Atento (2001–2002).

EDUCATION, TRAINING & LANGUAGES

B.S. Computer Engineering – UNIVESP (Brazil) [2019 – Present (In progress)]

Data Processing Technologist – FATEC-SP [2003 – 2007]

Certifications & Courses: Building Web Apps in Django (Univ. of Michigan) • Full Stack Developer Program (PUC-Rio)

Languages: English (C2 Proficient / Mastery) • Portuguese (Native)